

PROGRAM-1

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    char cmd[100];

    printf("Enter Linux command: ");
    scanf("%s", cmd);
    pid_t pid = fork();
    if(pid == 0)
    {
        printf("Child PID : %d\n", getpid());
        execlp(cmd, cmd, NULL);
        perror("Execution Failed");
    }
    else if(pid > 0)
    {
        printf("Parent PID : %d\n", getpid());
        wait(NULL);
        printf("Child process completed.\n");
    }
    else
    {
        printf("Fork Failed\n");
    }
    return 0;
}
```

OUTPUT -:

```

student@DESKTOP-UIMU3UD:~$ nano cmd_exec.c
student@DESKTOP-UIMU3UD:~$ gcc cmd_exec.c -o cmd_exec
student@DESKTOP-UIMU3UD:~$ ./cmd_exec
Enter Linux command: date
Parent PID : 31
Child PID : 32
Tue Aug 11 06:00:22 UTC 2026
Child process completed.

```

```

student@DESKTOP-UIMU3UD:~$ ls
Hello.c      a.txt      cmd_exec.c  file2      fork_exec2  fork_test.c  myshell  nano.25.save  parent_child_pi
d            pid.c      shortcut    wait_example.c
Hello.c.save b.txt      exec_example fork_exec   fork_exec2.c hello        name      parent_child  parent_child_pi
d.c          run_all.sh test.c      zombie
OSLab        cmd_exec   exec_example.c fork_exec.c fork_test   hello.c      name.c     parent_child.c pid
script.sh    wait_example zombie.c
student@DESKTOP-UIMU3UD:~$ pwd
/home/student
student@DESKTOP-UIMU3UD:~$ uname
Linux

```

```

student@DESKTOP-UIMU3UD:~$ lscpu
Architecture:          x86_64
  CPU op-mode(s):      32-bit, 64-bit
  Address sizes:        36 bits physical, 48 bits virtual
  Byte Order:           Little Endian
CPU(s):                4
  On-line CPU(s) list: 0-3
Vendor ID:              GenuineIntel
Model name:             Intel(R) Core(TM) i3-4005U CPU @ 1.70GHz
  CPU family:           6
  Model:                69
  Thread(s) per core:   2
  Core(s) per socket:   2
  Socket(s):            1
  Stepping:             1
  CPU(s) scaling MHz:   100%
  CPU max MHz:          1701.0000
  CPU min MHz:          0.0000
  BogoMIPS:             3402.00
  Flags:                fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov p
at pse36 clflush dts acpi mmx fxsr sse sse2 ss ht tm pbe syscall nx pdpe1gb rdtsc
  cp lm pni pclmulqdq dtes64 monitor ds_cpl vmx est tm2 ssse3 fma cx16 xt
pr pdcm pcid sse4_1 sse4_2 movbe popcnt tsc_deadline_timer aes xsave osx
  save avx f16c rdrand lahf_lm abm fsgsbase tsc_adjust bmi1 avx2 smep bmi2 erms i
nvpcid ibrs ibpb stibp ssbd
Virtualization features:
  Virtualization:       VT-x
  Hypervisor vendor:     Windows Subsystem for Linux
  Virtualization type:   container

```

```

student@DESKTOP-UIMU3UD:~$ lsblk
lsblk: failed to access sysfs directory: /sys/dev/block: No such file or directory
student@DESKTOP-UIMU3UD:~$ ps
  PID TTY          TIME CMD
   9 tty1          00:00:00 bash
  37 tty1          00:00:00 ps

```

PROGRAM-2

```

#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main()
{
    int src, dest;
    char buffer[1024];
    int bytes;

    src = open("input.txt", O_RDONLY);

    if(src < 0)
    {
        printf("Cannot open source file\n");
        return 1;
    }

    dest = open("output.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);

    while((bytes = read(src, buffer, sizeof(buffer))) > 0)
    {
        write(dest, buffer, bytes);
    }

    close(src);
    close(dest);

    printf("File copied successfully.\n");

    return 0;
}

```

OUTPUT -:

```

student@DESKTOP-UIMU3UD:~$ echo "Hello, this is test content for file copy." > input.txt
student@DESKTOP-UIMU3UD:~$ nano file_copy.c
student@DESKTOP-UIMU3UD:~$ gcc file_copy.c -o file_copy
student@DESKTOP-UIMU3UD:~$ ./file_copy
File copied successfully.
student@DESKTOP-UIMU3UD:~$ cat output.txt
Hello, this is test content for file copy.
student@DESKTOP-UIMU3UD:~$ █

```